

BOOKLET NUMBER: **52886**

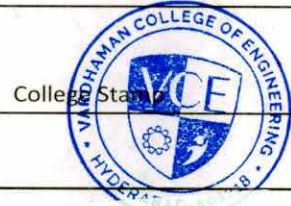


VARDHAMAN

COLLEGE OF ENGINEERING

Shamshabad – 501 218, Hyderabad

CAT EXAMINATION ANSWER BOOKLET



VARDHAMAN COLLEGE OF ENGINEERING
23881A05EC(G)
A8703 23.10.2025 AN



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CAT - I / II EXAMINATION I / II / III / IV B.Tech / M.Tech / MBA I / II Semester 10/2025

(Strike off whichever is not applicable)

(Month and Year)

Name : Sathvik Rudam
Branch : Computer Science and Engineering
Subject : Machine Learning
Subject Code : A8703
Date of Exam : 23-10-2025

Signature of the Invigilator with date

[Signature]
23.10.25

(Booklet contains 20 pages)

INSTRUCTIONS TO THE CANDIDATES

- This booklet contains 20 pages. Candidates must ensure it before writing and in case a defective answer book is issued it must be returned to the invigilator and a new and defect free booklet must be obtained.
- Before the candidate begins to answer, registered number, particulars of year, semester, subject, etc., are to be filled in. Failure to enter all or any of these particulars may disqualify the paper from valuation.
- Candidate is prohibited from
 - writing:
 - anything addressing the examiner in any manner whatsoever, in their answer book.
 - objectionable/obscene language in the answer book.
 - anything other than their Registered No. on the question paper.
 - either seeking or providing any assistance to the fellow candidates in the exam.
 - possessing a manuscript or a printed matter, in any form, in the examination hall.
 - bringing loose sheets or paper into the examination hall and detaching any paper from the answer book.
 - carrying Mobile Phone, Smart Watches and any other Electronic Gadgets etc., to Exam Hall.

Violation of these instructions will be viewed as a case of malpractice, which is a punishable offence.

- Before beginning to answer any question, candidates must write the correct question number, and should not write
- Answers must be written legibly on both sides of the paper. There shall be about 25 lines in each page. It is not necessary to begin each answer on a fresh page. Candidates should not use any other ink, except BLACK or BLUE ink.
- Rough work, if any, must be separated, from the subject matter, by a line and noted as rough work.
- The answer book, at the end of the examination, must be handed over to the Assistant Superintendent (Invigilator) by the candidate.

This responsibility lies with the candidate only.

- Candidates should maintain absolute silence during the time of examination. Misbehavior, in any form, by the candidate, in the examination hall, will attract severe punishment.
- Candidates are permitted to leave the examination hall only after the allotted time.
- No additional answer books will be supplied.

To be filled in by the Examiner only

MARKS AWARDED

MARKS AWARDED										
PART – A				PART – B						
Q No.1				Q No.	2	3	4	5	6	7
a		f		a						
b		g		b						
c		h		c						
d		i		d						
e		j		Total						
Total (a to j)										
In Words:										
In Figures:										

Signature of the Examiner

START WRITING FROM HERE

Q.No.

PART-A

1) a) → likelihood probability refers to the term in Naive Bayes Classifier.

$$P(A/B) = \frac{P(A) P(B/A)}{P(B)}$$

where: $P(A)$ = likelihood Probability

$P(B/A)$ = Pre-probability

$P(A/B)$ = Post probability

$P(B)$ = marginal likelihood.

→ likelihood probability means the likelihood occurrence of a required event wrt another event.

Ex: Probability of a student failing in Data Science who has failed in AI. Here, student failing in DS is likelihood probability.

1) b) → Two applications of Naive Bayes Classifier are:

i) Spam Mail Detection: Almost every mail providing app is inbuilt with spam detector using Naive Bayes Classifier.

ii) Disease Detection: IBM has developed an ML model which can detect patients with Breast Cancer.

iii) Weather Forecasting: Weather Forecasting can also be done using Naive Bayes Classifier.

Q.No.

i) c) KNN is called lazy learner algorithm because it doesn't learn anything from the dataset.

→ KNN algo just stores the dataset and directly starts measuring distance with test points. As it doesn't learn anything, it is called as lazy learning Algorithm.

i) d) → Cluster Analysis is one of the types of Unsupervised Learning methods. It refers to grouping of similar type of data into one cluster (group).

→ Cluster analysis helps segregate data into different groups of similar items. It can either be used as preprocessing technique or independently to group data.

Ex: Clustering of apples, bananas, from a basket of fruits.

i) e) → Partitioning methods in clustering are K-means, and K-medoids.

→ Partitioning methods work by choosing K-no. of points (or) data as center points, and segregate the other points by putting them into cluster whose point (centre point) is close to them. This point is repeated until all the data is correctly clustered.

i) f) The objective function of K-Means algorithm is to segregate different items by clusters where each cluster has

i) maximum intra-cluster similarities

ii) minimum inter-cluster similarities

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1) g) → Synaptic weights are added to Artificial neurons to ~~put~~ affects their ~~you~~ output.

→ Synaptic weights are two types:

i) excitatory (or) positive weights

ii) inhibitory (or) negative weights.

positive weights results in the output function, whereas negative weight results in negative output function.

1) h) → ADALINE stands for Adaptive Linear Neural Network.

→ ADALINE is a type model of Neuron

→ It only has two outputs.

→ The output depends on positive and negative I/Ps.

1) i) Hyperbolic Tangent function

$$f(x) = y_{out} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

→ Hyperbolic Tangent function is 5th type of activation function.

→ It is differentiable.

→ The other four activation functions are:

i) identity function

ii) Step/Threshold function

iii) Rectified Linear Unit function

iv) Sigmoid function: a) Binary Sigmoid

b) Bipolar Sigmoid

Q.No.

- 1) j) → K-means clustering is not suitable for large data, as it takes many iterations to find the best clusters.
→ Choosing the value of k is difficult
→ It is time taking process as it needs to perform many iterations.

PART-B

2) Given:

Predict win probability when
win = yes, Weather Conditions = Overcast,
wins in last three matches = 1
Humidity = normal
Toss = won.

First, let us find probability of won matches

$$P(\text{Overcast}) P(\text{Won} | \text{Overcast}) = 9/14$$

$$P(\text{Won} | \text{No}) = 5/14$$

Now,

$$P(\text{Overcast} | \text{yes}) = 4/9, \quad P(\text{Overcast} | \text{no}) = 0/5$$

$$P(\text{last 3} | \text{yes}) = 3/9, \quad P(\text{last 3} | \text{no}) = 1/5$$

$$P(\text{Humidity} | \text{yes}) = 6/9, \quad P(\text{Humidity} | \text{no}) = 1/5$$

$$P(\text{Toss} | \text{yes}) = 3/9, \quad P(\text{Toss} | \text{no}) = 3/5$$

Now, according to Bayes theorem,

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$$\begin{aligned} & \text{Prob (winning match, overcast, won 1 of last 3, Normal, toss)} \\ &= P(\text{Overcast}_{\text{yes}}) \times P(1 \text{ last 3 matches}) \times P(\text{Humidity}_{\text{yes}}) \times P(\text{Toss}_{\text{yes}}) \times P(\text{Yes}) \\ &= \frac{2}{9} \times \frac{3}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{1}{14} = \frac{4}{27 \times 7} = 0.02 \end{aligned}$$

$$\begin{aligned} & \text{Prob (Losing Match)} = \\ & P(\text{Overcast}_{\text{No}}) \times P(1 \text{ last 3}_{\text{No}}) \times P(\text{Humidity}_{\text{No}}) \times P(\text{Toss}_{\text{No}}) \times P(\text{No}) \\ &= 0 \end{aligned}$$

Q.No.

3) → Random Forest Algorithm is a Classification algorithm.

→ It is a combination of multiple decision trees in simple words.

→ Random forest Algorithm uses Bagging concept on multiple sub trees, and performs hard voting. In this way, it chooses best tree.

→ Algorithm:

i) Consider 'm' ($m < n$) no. of subsets where 'n' is the total no. of dataset values. Consider 'm' values randomly even their records too.

ii) Now, perform decision tree algorithm by using these subsets. Find the max depth of the tree using best tree method.

iii) Construct decision tree i.e. split the data by calculating entropy and Info gain. Consider the highest Info. gain attribute, and split the dataset into two parts.

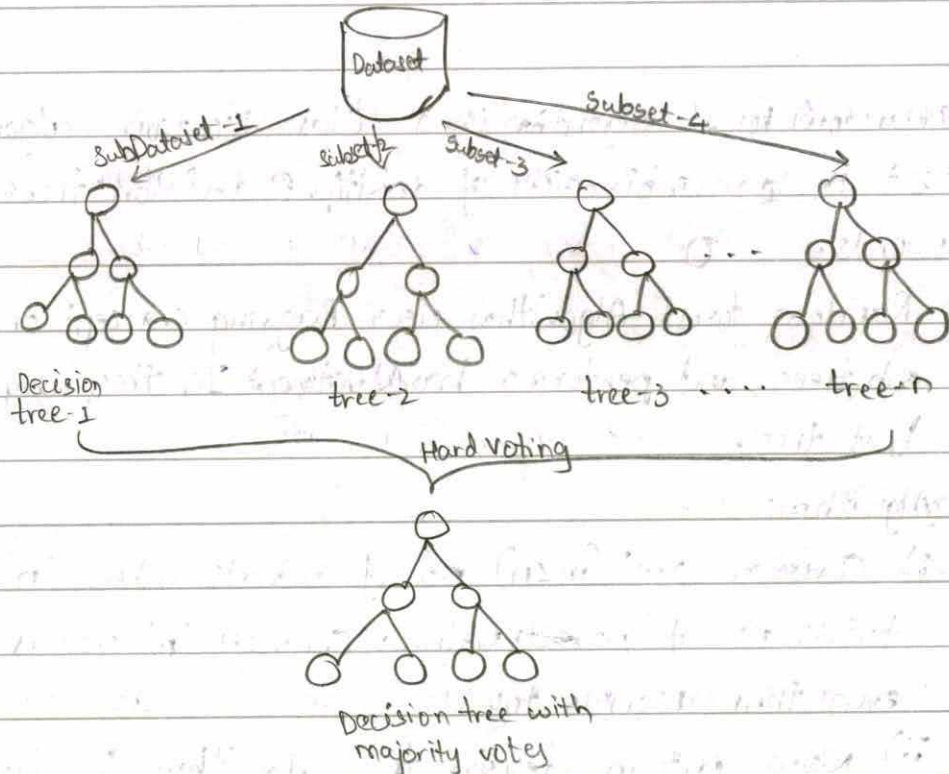
iv) Now, repeat this step until the tree reaches it's end.

v) Now, take different dataset and perform steps 2 to 4 until we get many decision tree.

vi) Now, using hard voting process, we select the decision tree.

→ As we are taking decision tree from multiple decision trees, hence the name Random Forest.

Q.No.



- Here, we first take dataset and divide it into multiple subsets.
- Now, consider multiple decision trees and apply/perform bagging on all trees, using data from same dataset i.e. different subsets.
- The obtained decision trees, Hard voting is performed.
- The majority tree wins.
- Advantages
 - More efficient as it is combination of multiple random trees.
 - Doesn't get turned to bias, whereas Random for decision tree gets biased for large data.
 - The trees which are unused after voting can be used for testing.

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Disadvantage:

→ As it is a combination process of multiple decision trees, it is a bit high space taking.

→ Bagging refers to sharing/applying same data on multiple models and combining the results. This is done in Random Forest.

Applications:

→ Used to detect diseases.

→ Used to predict prices, stocks, loan frauds, etc.

→ Random Forest is both Classification and Regression Algorithm

Q.No.

4) $\rightarrow$ k-Medoids is a Partitioning Clustering Algorithm.

$\rightarrow$ In k-Medoids, distance between centroid and a point is found using the formula $|x_i - x_j|$.

$\rightarrow$ The cost is calculated for a cluster as

$$\sum |p_i - x_j|$$

where p_i = cluster point

x_j = Point of Test (or) Test point.

$\rightarrow$ Here, we first consider the dataset, and consider k number of points as medoids, i.e. centroids.

$\rightarrow$ Now, we calculate distance b/w all the test points to medoids, and classify them as clusters.

$\rightarrow$ Now, calculate the cost of cluster using

$$\text{cost} = \sum |p_i - x_j|$$

$\rightarrow$ Choose, any non-medoid point and replace it with 2nd medoid point.

$\rightarrow$ Now, perform clusterising again and calculate cost

$\rightarrow$ If obtained cost is less than old cost, replace the new cost as medoid cost.

$\rightarrow$ Else, let the old clusters be medoid clusters.

Algorithm:

i) Consider k-no. of points randomly as medoids

ii) Now, clusterise the dataset based on shortest distance to medoid.

iii) Calculate the cost of each cluster & save in cost

iv) Replace second medoid point with one non-medoid point.

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iv) Repeat steps 2 and 3.

vii) If $\text{Cost}_{\text{new}} < \text{Cost}$:

then $\text{Cost} = \text{Cost}_{\text{new}}$

New medoids are fixed.

else

change to old medoids.

Advantages:

→ It is very simple algorithm, easy to understand.

→ In this, the medoid point will be a point of dataset.

Disadvantages:

→ It takes very time to perform classification.

Ex: consider dataset $\{2, 3, 5, 12, 6\}$.

Now, I want to divide it into 2 clusters. Then, take two medoids, assume 2 & 6.

Now, perform distance calculation. If the question If I want to measure using Manhattan distance, then

Point	M_1	M_2	Cluster
2	0	4	C_1
3	1	3	C_1
5	3	1	C_2
12	10	6	C_2
6	4	0	C_2

Distance = $|x_1 - x_2|$

$M_1 = 2$

$M_2 = 6$

→ Now, compute cost $\Rightarrow 0 + 1 + 3 + 10 + 4 + 4 + 3 + 1 + 6 + 0 = 32$

→ Consider a new medoid point randomly & replace.

Q.No.	Let us consider '5', then $M_1=2$, $M_2=5$			
	Point	M_1	M_2	Cluster
	2	0	3	C_1
	3	1	2	C_1
	5	3	0	C_2
	12	10	7	C_2
	6	4	1	C_2

Now, cost = $1+3+10+4+3+2+7+1 = 31$

So, medoids are 2,5

Repeat process until, shortest cost found.

5) Given data.

60, 75, 90, 105, 120, 135, 150, 165, 180, 195

centroids: $C_1 = 75$ (centre 1)

$C_2 = 120$ (centre 2)

$C_3 = 165$ (centre 3)

⇒ Data C_1 C_2 C_3 Cluster.

60	15	60	105	C_1
75	0	45	90	C_1
90	15	30	75	C_1
105	30	15	60	C_2
120	45	0	45	C_2
135	60	15	30	C_2
150	75	30	15	C_3
165	90	45	0	C_3
180	105	60	15	C_3
195	120	75	30	C_3

Q.No.

Now, clusters are: $C_1: \{60, 75, 90\}$

$$C_2 = \{105, 120, 135\}$$

$$C_3 = \{150, 165, 180, 195\}$$

perform avg of each cluster:

$$C_1 = \frac{60+75+90}{3} = \frac{225}{3} = 75$$

$$C_2 = \frac{105+120+135}{3} = \frac{360}{3} = 120$$

$$C_3 = \frac{150+165+180+195}{4} = \frac{690}{4} = 172.5$$

$\Rightarrow$ New points obtained, perform again.

Data	C_1	C_2	C_3	Cluster
60	15	60	112.5	C_1
75	0	45	97.5	C_1
90	15	30	82.5	C_1
105	30	15	67.5	C_2
120	45	0	52.5	C_2
135	60	15	37.5	C_2
150	75	30	22.5	C_3
165	90	45	7.5	C_3
180	105	60	7.5	C_3
195	120	75	22.5	C_3

As we can see, the cluster obtained are

$$C_1: \{60, 75, 90\}$$

$$C_2: \{105, 120, 135\}$$

$$C_3: \{150, 165, 180, 195\}$$

Q.No.

→ As there is no change, these are the final clusters.

7) → Architecture of Feed-Forward Neural Nlw:

→ Four types of Architectures exist:

- i) Single Layer Feed-Forward NN
- ii) Multi-Layer Feed Forward NN
- iii) Competitive Feed Forward NN
- iv) Repetitive Feed Forward NN

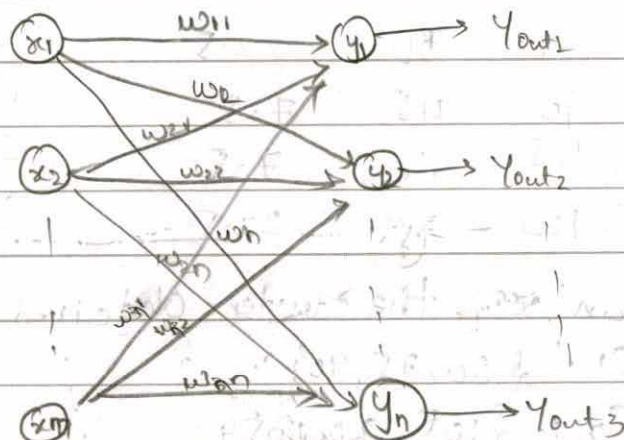
i) Single Layer Feed Forward NN:

→ Only two layers are present in this NN.

→ The calculations are performed on the OP layer.

→ The data is directly passed from IP layer to OP layer. Hence

→ As the calculation is performed only one one layer, it is called Single layer feed forward.



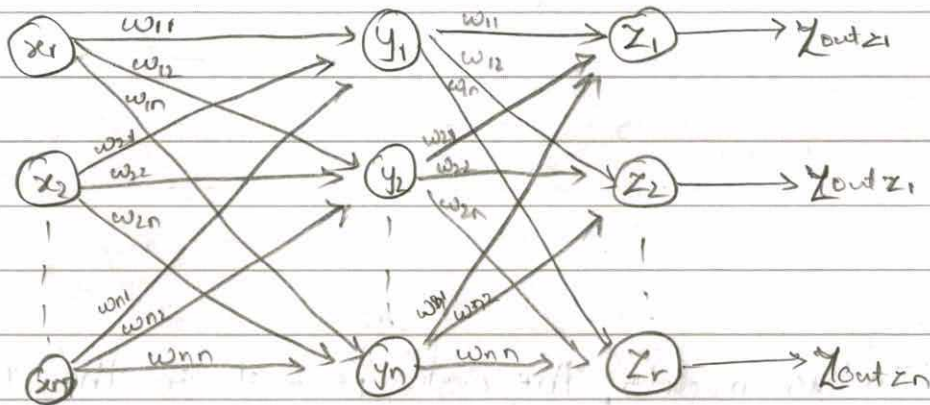
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ii) Multi-Layer Feed Forward NN:

→ Here, it is same as single layer feed forward NN. But,

→ the only difference is there may exist one or more intermediate layers.

→ Calculations are performed on layers including Output (Op) layer and middle layer.

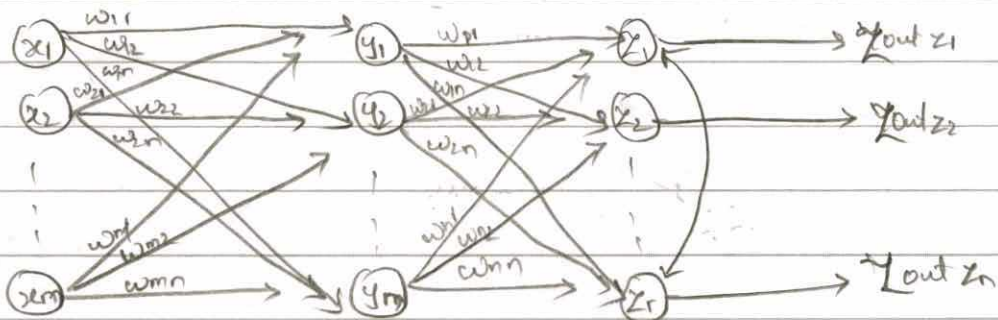


iii) Competitive Feed Forward NN:

→ This ~~for~~ Neural Network is same as Multi-layer

→ But, there is a interlink between output layers

→ The output layers compete with each other to represent the input



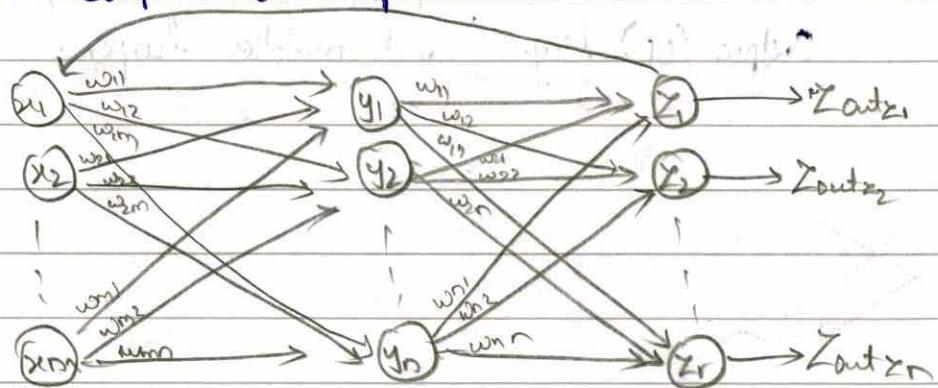
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iv) Repetitive Neural Network:

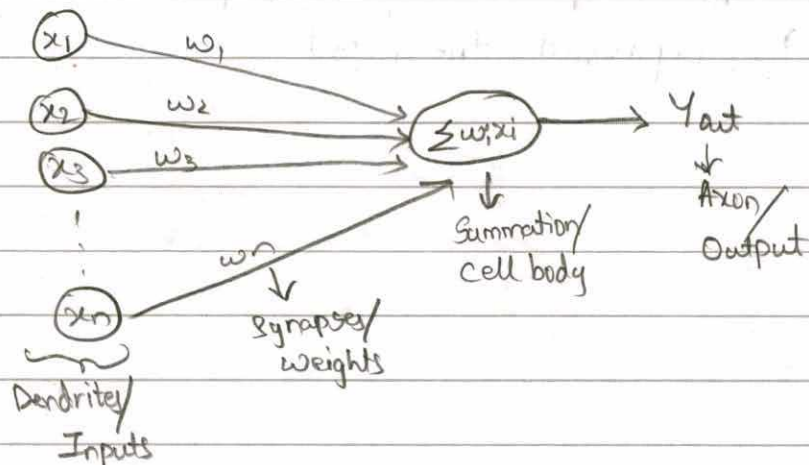
→ Here, it is same as Multi layer NN.

→ But, here exists one special thing.

→ As every architecture is forward moving, But in this model, there exists a feedback loop from Output to Input.



→ In this models, the data is sent to the Input layer, then the data is passed through hidden layers by performing calculations, and reaches Output layer. It is then passed to next Neuron



$$\begin{array}{r} 189 \\ \times 2 \\ \hline 378 \end{array}$$

$$27 \times 7$$

$$\begin{array}{r} 189^4 \\ \hline 4 \end{array}$$

Q.No.

$$\begin{array}{r} 6 \text{ (13)} \\ 173 \\ -105 \\ \hline 68 \end{array}$$

$$\begin{array}{r} 6 \text{ (13)} \\ 173 \\ -135 \\ \hline 38 \end{array}$$

$$\begin{array}{r} 189 \text{) } 4 \text{ (0.02} \\ \underline{0} \\ 400 \end{array}$$

$$\begin{array}{r} 173 \\ -150 \\ \hline \end{array}$$